Optimizing UWB CIR Truncation for NLOS Identification on MCU Deployment

Yuqing Jiang, Hanbo Liu, Zhangyi Yang, Jiaxin Wu, and Wei Dong

***Abstract*— Truncating the channel impulse response (CIR) is a critical step for non-line-of-sight (NLOS) identification within ultra-wideband (UWB) systems to manage data complexity. This letter formulates the process as a constrained and quantifiable optimization problem, thereby achieving an optimal balance between computational performance and classification accuracy. To implement this methodology, we first designed a lightweight attention-assisted network to serve as an effective evaluation tool. Leveraging this network, we then conducted a systematic search that identified an optimal input configuration: a CIR subset containing only 63 sample points. This co-optimized input model pair achieves a remarkable NLOS recall of 93.68%. More importantly, this result provides a critical a priori determination for deploying high precision localization algorithms on resource constrained MCUs. This co-optimized system has been successfully quantized and deployed on an STM32 MCU, verifying its high efficiency and feasibility for industrial grade applications.**

***Index Terms*— UWB, NLOS classification, CIR truncation, TinyML, resource-constrained devices.**

I. Introduction

ACCURATE Non-Line-of-Sight (NLOS)/Line-of-Sight (LOS) identification is a cornerstone for implementing reliable Ultra-wideband (UWB) positioning systems [1], [2], [3]. With the advancement of deep learning [4], [5], [6], [7], [8], the field currently faces a profound trade off among model accuracy, data complexity, and the practical deployment capabilities of MCUs, leading to the emergence of several distinct research paradigms.

The paradigm pursuing ultimate generalization and performance, represented by models like RIM-Net [9], aims for the highest classification accuracy by processing complete, complex channel impulse response (CIR) information. However, their large parameter count makes Microcontroller Unit (MCU) deployment infeasible. Another paradigm, exemplified by Compensated Ranging Weighted Least Square, completely forgoes CIR data, relying solely on easily accessible scalar information such as ranging and received signal strength (RSS) [10]. This approach maximizes hardware compatibility but sacrifices the rich discriminative information within the CIR, thereby limiting the upper bound of its classification accuracy. Although incorporating auxiliary sensors [11] enhances ranging precision, it inevitably increases hardware costs.

For TinyML [12] applications aimed at high energy efficiency, the current state-of-the-art (SOTA) [13] approach involves selecting approximately 50 CIR sample points near the First Path (FP) and combining them with 7 features to construct a low dimensional input vector, achieving excellent performance on MCUs. However, these works share a fundamental limitation: the selection of the input data subset remains largely dependent on the engineer's intuition. There is no guiding theoretical or practical basis for this choice.

Furthermore, the performance of advanced UWB localization algorithms [14], [15], such as Weighted Least Squares (WLS) [10], [16] or Extended Kalman Filter (EKF) [17], [18], [19], heavily relies on an accurate a priori judgment of the quality of each range measurement. Therefore, a high accuracy, high efficiency NLOS/LOS classifier that can run on an MCU is a key component for deploying these advanced algorithms on real hardware.

This letter proposes a systematic, data driven approach for CIR truncation. We achieve this by first designing a lightweight hybrid network tailored for on device deployment. We then utilize this network to conduct search, resulting in a co-optimized solution that is not only high performing but also methodologically instructive and directly deployable on resource-constrained devices.

II. Algorithm Design

A. Network Design

The architecture of our proposed network, depicted in Fig. 1, is a parallel dual channel model designed to process both CIR time-series data and auxiliary physical parameters. Key hyperparameters are detailed in Table I. The data flows through the network as follows:

The CIR sequence is fed into a CNN Feature Extractor. This module features a multi-kernel residual architecture to extract multi-scale temporal features. As detailed in Table I, the architecture incorporates two Max-pooling layers, which effectively compress the time series length to 1/4 of the input. The resulting feature sequence is then passed to a Self-Attention Module, which weighs the importance of different time steps to focus on the most discriminative waveform segments. Global Average Pooling is utilized as a residual connection to mitigate the gradient vanishing problem.

Concurrently, the 8 auxiliary parameters are processed by a simple MLP block, labeled Params Mapping in Fig. 1.

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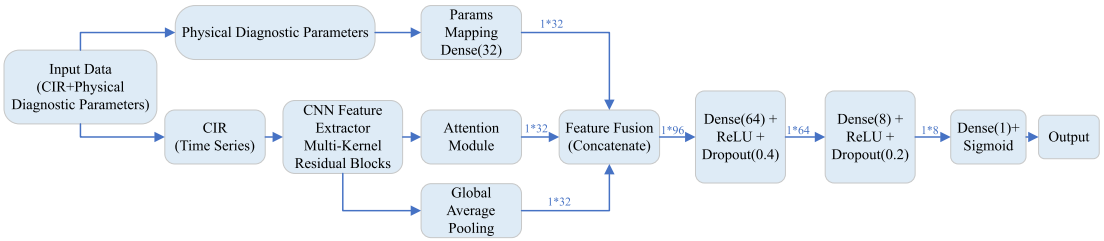


Fig. 1. Network design.

TABLE I
MODEL PARAMETER

Module	Parameter	Value
Data Input	CIR Length	63 (Variable)
	Num of Extra Params	8
CNN Feature Extractor		
Block 1: Multi-Scale Input	Num of Filters	4
	Size of Kernel	[3,5,7,9,11,13,15,17]
Block 2/3: Deep Feature Extraction	Kernel Size	Block 2: 7 Block 3: 5
	MaxPooling1D Layer	Pool / Strides: 2 / 2
Self-Attention Module	Number of Heads	8
	Model Dimension	32
Fusion & Classifier Head	Num of Layer	3
	Neurons	64 / 8 / 1

Fusion and Classification: The Feature Fusion block concatenates the outputs from the CNN branch, the Self-Attention Module, and the Parameter Branch. Finally, this fused feature vector is fed into a Classifier (MLP) head to produce the final prediction.

To ensure robust training, standard techniques like Batch Normalization and Dropout are utilized throughout the network. Specifically, a dropout rate of 0.4 is applied after the 64-neuron layer, while a lower rate of 0.2 is used following the 8-neuron layer.

B. Feature Selection and Preprocessing

To reduce the computational burden and enhance model efficiency, data preprocessing is a critical step. The model's input consists of two parts: the processed CIR time-series sequence and the physical diagnostic parameters. To determine the optimal length of the effective CIR segment, it is necessary to model and characterize the statistical properties of the CIR [20], [21]:

$$h_{\text{discr}}(t) = \sum_{l=0}^{L} \sum_{k=0}^{K} a_{k,l} \exp(j\phi_{k,l}) \delta(t - T_l - \tau_{k,l}) \quad (1)$$

where $a_{k,l}$ is the tap weight of the kth component in the lth cluster, T_l is the delay of the lth cluster, and $\tau_{k,l}$ is the delay of the kth MPC relative to the lth cluster arrival time T_l. The phases $\phi_{k,l}$ are uniformly distributed, i.e., for a bandpass system, the phase is taken as a uniformly distributed random variable from the range $[0, 2\pi]$.

Analyzing the statistical characteristics of the CIR within the environment, the analysis of the Power Delay Profile (PDP) collected under NLOS conditions yields:

$$E\left\{|a_{k,l}|^2\right\} \propto (1 - \chi \cdot \exp(-\tau_{k,l}/\gamma_{\text{rise}})) \cdot \exp(-\tau_{k,l}/\gamma_l) \quad (2)$$

where χ describes the attenuation of the first path component, γ_{rise} determines the rate at which the PDP rises to its local peak, and γ_l describes the subsequent decay rate.

In indoor environments, the rise time constant γ_{rise} typically exceeds 15 ns. Consequently, under NLOS conditions, the CIR requires a significant duration to rise from the noise floor to its peak. This indicates that the direct path arrives at the receiver well before the signal peak is reached. Commercial chips, such as the DW1000, typically employ a noise threshold-based mechanism to detect the FP.

$$Threshold = STD_NOISE \times NTM \quad (3)$$

where the Noise Threshold Multiplier (NTM) is generally set to 12 or 13 [22]. STD_NOISE represents the standard deviation of noise estimated by the front-end.

During online system operation, the truth FP is unavailable. Therefore, we utilize the estimated FP provided by the LDE algorithm, denoted as FP_{Predict}. When preprocessing data for the neural network input, we similarly use FP_{Predict} as the reference anchor to truncate the effective CIR segment. We look backward to capture the signal during the energy rising phase and look forward to utilize the multipath components.

Let $r[n]$ denote the amplitude sequence of the CIR provided by the digital receiver. In this work, we define two parameters to truncate the CIR: the look-back window size X and the total sequence window size Y. Therefore, the multipath window size is $Y - X$. The truncated sequence s is defined such that its ith element is given by:

$$s_i = r[FP_{\text{Predict}} - X + i], \quad i = 0, 1, \ldots, Y - 1 \quad (4)$$

Fig. 2 illustrates the average signal profile preceding FP_{Predict}. The energy begins to diverge from the noise threshold at approximately index -3 for LOS conditions and index -10 for NLOS conditions. Consequently, we define the search range for the look-back window X as $3 \leq X \leq 10$.

Fig. 3 depicts the average signal profile following FP_{Predict}. The signal intensity drops to -20 dB at index 42 under LOS conditions, whereas under NLOS conditions, this decay point

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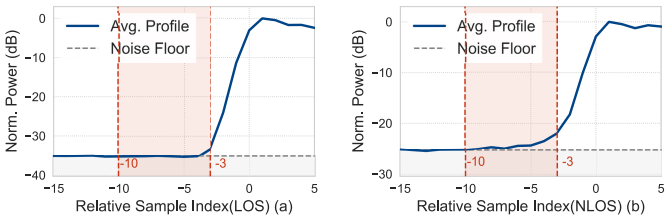


Fig. 2. Average signal profile preceding the FP_{Predict}.

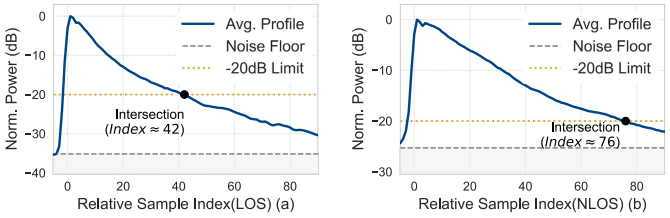


Fig. 3. Average signal profile preceding the FP_{Predict}.

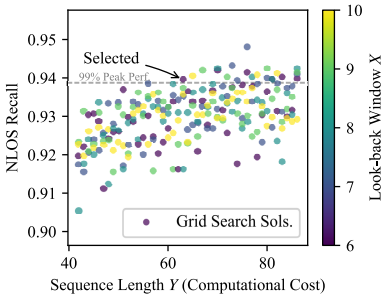


Fig. 4. Optimizing CIR truncation parameters.

extends to index 76. Therefore, we set the search range for the multipath window size $(Y - X)$ as $42 \leq (Y - X) \leq 76$.

The DW1000 reports the CIR at intervals of 1 ns. Given that electromagnetic waves propagate at the speed of light, the interval between two adjacent CIR samples corresponds to a spatial distance of approximately 0.3 m. In the positioning subset of the dataset, 99% of the ranging errors under NLOS conditions fall within 1.8 m [23]. This implies that the true FP is located in the range from $FP_{\text{Predict}} - 6$ to FP_{Predict}. To ensure coverage of the true FP, we imposed an additional constraint on the look-back window selection, refining the search range to $6 \leq X \leq 10$.

We performed a search for the optimal parameter combination within the ranges $6 \leq X \leq 10$ and $42 \leq Y \leq 86$, utilizing the model's NLOS Recall on the validation set as the primary performance metric. We defined a performance threshold at 99% of the global maximum recall achieved within this search space. From the candidates exceeding this threshold, we selected the combination with the minimum Y representing the lowest computational cost as the optimal solution. This approach effectively minimizes computational overhead while maintaining near-peak performance, fulfilling the requirements for deployment on resource-constrained edge devices. The results of this selection strategy are illustrated in Fig. 4. Based on this analysis, we selected $X = 6$, $Y = 63$ as the final CIR truncation parameters.

Physical Diagnostic Parameters: In addition to the CIR sequence, UWB chips provide a series of physical diagnostic

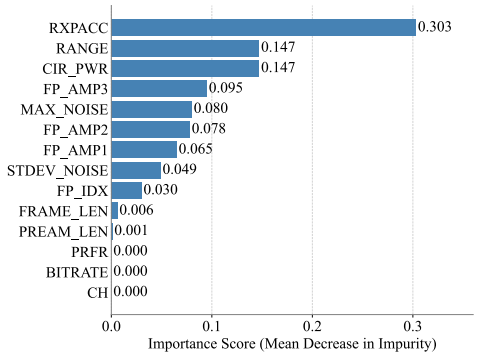


Fig. 5. Optimizing CIR truncation parameters.

parameters that reflect channel quality. As shown in Fig. 5, we used Random Forest and Recursive Feature Elimination to assess the importance of each parameter, ultimately selecting the 8 features most relevant to NLOS phenomena as auxiliary inputs for the model. As shown in Fig. 5, we selected the top 8 features based on their importance scores to serve as auxiliary inputs for the model.

III. Dataset and Experiment Design

A. Dataset and Partitioning

All experiments use the public eWINE [23] dataset. We leverage its Positioning Subset, which contains ground truth ranging data, for the error analysis in Section II-B. For model training and evaluation, we use the larger and more balanced Classification Subset. We employ a consistent 6:2:2 data split (train/validate/test) across all experiments. The validation set is utilized for model selection and parameter optimization, while the results reported in the ablation studies and comparative evaluations represent the selected model's performance on the independent test set.

B. Implementation Details

Edge Platform: The model was deployed and verified on the target edge device. The core of this device is an STM32F405RG MCU which features 1MB of Flash and 192KB of RAM. We used the X-CUBE-AI toolchain to convert the model into C code for deployment on the MCU.

Model Training Configuration: The model was trained in TensorFlow 2.10.1. We used the Adam optimizer with Focal Loss $(\gamma = 2, \alpha = 0.8)$, a learning rate of 0.002, and a batch size of 256 for 200 epochs.

Quantization Configuration: The 8-bit integer (INT8) quantized model was generated using a post-training full integer quantization (PTQ) strategy. During the quantization process, a representative dataset was constructed by randomly sampling 15 batches from the training set to accurately calibrate the dynamic ranges of weights and activations.

C. Evaluation Metrics and Baselines

Evaluation Metrics: We used four standard classification metrics: Accuracy, Precision, Recall, and F1-Score. In UWB positioning, misclassifying an NLOS signal as LOS leads to more severe errors, making NLOS Recall the most critical metric for practical value.

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TABLE II
PERFORMANCE OF DIFFERENT MODEL

Method	Accuracy	Precision	Recall	F1-score
CNN [18]	87.4%	85.9%	89.4%	87.6%
CNN-LSTM [24]	82.37%	79.44%	87.36%	83.21%
FCN-Attention [25]	88.24%	85.85%	91.56%	88.62%
Self-Attention MLP (Unquantized) [12]	93.10%	93.21%	93.13%	93.17%
Self-Attention MLP (Quantized) [12]	92.99%	92.65%	93.42%	93.03%
Our Model ($X = 2, Y = 50$) (Unquantized)	92.65%	92.99%	92.36%	92.67%
Ours (Unquantized)	**93.10%**	**92.38%**	**94.04%**	**93.20%**
Ours (Quantized)	**91.94%**	**90.62%**	**93.68%**	**92.12%**

TABLE III
ABLATION STUDY RESULTS

Method	Accuracy	Precision	Recall	F1-score
Only CNN	79.79%	86.81%	70.54%	77.83%
Only MLP	91.03%	93.27%	88.57%	90.86%
CNN + Self-Attention	80.24%	81.79%	78.40%	80.06%
CNN + MLP	92.94%	93.53%	92.36%	92.94%
Ours (Unquantized)	**93.10%**	**92.38%**	**94.04%**	**93.20%**

TABLE IV
DEPLOYMENT TEST

Method	RAM (KB)	Flash (KB)	Time (ms)
Self-Attention MLP (quantized)	10.3	—	< 1
Our Model ($X = 2, Y = 50$) (Unquantized)	19.24	93.80	34.5
Ours (Unquantized)	24.22	93.82	44.1
Ours (Quantized)	15.79	24.11	29.0

Baselines: We compared our model against several representative existing methods. To explicitly evaluate the contribution of our optimization strategy, we conduct a controlled comparison using our model paired with the CIR input configuration $(X = 2, Y = 50)$ employed by the SOTA baseline.

Additionally, ablation studies were conducted by removing the self-attention module or the MLP branch to verify the effectiveness of each component.

IV. EXPERIMENT ANALYSIS

A. Performance

We compare the performance of our proposed model, utilizing the optimized CIR sequence, against previously reported SOTA models. As shown in Table II, our model's performance on the key metric NLOS recall is comparable to that of SOTA, validating its competitiveness as a SOTA-level solution. Compared to the truncation parameters employed by the SOTA baseline, our optimized configuration improves the NLOS recall by 1.68%.

Unlike complex SOTA methods that rely on pre-training and fine tuning, our simpler alternative path focuses on systematically optimizing the input data itself. Our model exhibits excellent stability after quantization, with its recall metric decreasing by only 0.36%.

B. Ablation Study Analysis

To investigate and quantify the actual contributions of each key component within our model, we conducted a series of ablation studies.

As shown in Table III, the model performance is suboptimal when relying solely on either the CNN or MLP. However, fusing these branches yields a Recall of 92.36%, and incorporating the self-attention mechanism further improves the Recall to 94.04%.

C. Deployment

To validate its engineering value, we quantized and deployed the model on the STM32. Table IV showcases its resource efficiency: after INT8 quantization, the RAM footprint is merely 15.79 KB, with an inference latency of 29 ms. Given that a 1 Hz update rate is standard for many UWB positioning scenarios and computational latencies under 50 ms are generally negligible.

V. CONCLUSION

This letter addresses the critical challenge of deploying high performance UWB NLOS identification algorithms on resource constrained hardware. We successfully designed an attention assisted lightweight hybrid network and, more importantly, introduced a systematic, data driven methodology to optimize the input CIR truncation, replacing common heuristic approaches.

Leveraging the cumulative signal energy profile and the ranging error distribution, we derived the upper and lower bounds for both the CIR look-back and multipath windows. Subsequently, we employed our proposed parameter optimization strategy to select the optimal parameter combination for CIR truncation.

Finally, we validated the engineering value of our approach by successfully quantizing and deploying the optimized model onto an industrial grade STM32 microcontroller. The system exhibits excellent on device performance, requiring only 15.79 KB of RAM and achieving a 29 ms inference latency, while maintaining a high NLOS recall rate of 93.68%. This work provides a validated, efficient, and methodologically sound pathway for implementing advanced UWB positioning systems on real world edge devices.

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